A PROJECT TITLE

ON

Audience Engagement Classification Using Decision Trees and Random Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSE

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1. **Abstract:**

Measuring and understanding audience engagement is crucial across digital media, educational platforms, and live presentations. Traditional feedback mechanisms often fail to capture real-time, objective interest levels. This project proposes a machine learning-based approach to classify levels of audience engagement using Decision Trees and Random Forest algorithms.

By extracting behavioral, visual, or interaction-based features, the pipeline preprocesses relevant engagement indicators and constructs classification models. The Decision Tree model establishes an intuitive, rule-based baseline for engagement detection, while the Random Forest ensemble improves predictive accuracy, handles feature correlations, and reduces overfitting. Comparative evaluations demonstrate the performance of both models across metrics such as precision and overall accuracy. The resulting framework provides a reliable, scalable foundation for real-time engagement analytics and automated feedback systems.

Introduction:

Audience engagement plays an important role in understanding how effectively people interact with digital content, educational activities, and live presentations.
With the growth of digital platforms, analyzing audience behavior using machine learning has become increasingly useful.
This project focuses on classifying audience engagement levels using behavioral, visual, and interaction-based features.
Decision Tree and Random Forest algorithms are applied to identify engagement patterns and make accurate classifications.

The performance of both models is compared to determine an effective approach for reliable and automated audience engagement analysis.

Software Requirements:

Operating System: Windows / Linux / macOS
Programming Language: Python 3.x
IDE / Environment: Jupyter Notebook / Google Colab / VS Code
Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn
Dataset Format: CSV (audience engagement/behavioral data)
Machine Learning Framework: scikit-learn
Version Control: Git and GitHub

Signature of the Guide **Course Instructor**